

Joel Osian Roberts

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TECHNICAL SKILLS

Programming: C, C++, Python, Java, C#, PHP, Javascript, Powershell

Systems & Low-Level: STM32, Arduino, Microcontrollers, UART, I2C, State Machine, Interrupts, Memory Management, Debugging

Software Engineering: Object-Oriented Programming, REST APIs, Backend Development, System Design and Optimisation, Project Management, Requirements Gathering, User Acceptance Testing

Tools & Technologies: Git, Jira, Linux, Microsoft Power Platform, Automation Workflows, AI Enterprise Integration, Agile, AI Agent Development

EDUCATION

Bangor University, Bangor, Gwynedd, UK | BSc Software Engineering 2024-2027

- Related Coursework: Computer Systems, Middleware, Cybersecurity, Object-Oriented Programming

Grwp Llandrillo Menai, Pwllheli, Gwynedd, UK | A Levels 2022-2024

- Mathematics: A, Computer Science: A, Business: B
- Related Coursework: Data Structures, Computer Architecture, Mechanics & Statistics

EXPERIENCE

Adra

IT Degree Apprentice

Bangor, Gwynedd, UK

Aug 2024 - Present

Degree apprenticeship combining full-time professional experience with BSc study through weekly day-release and evening learning.

- Developed multiple internal applications, including a vehicle CO2 emissions tracking platform managing 400+ assets, integrating external REST APIs (DVLA) via HTTP requests and supporting real-time data processing and reporting, supporting corporate social responsibility.
- Engineered a secure and scalable document management portal supporting 1000s of external users, implementing validation, access control, and sensitive backend data handling.
- Contributed to a machine learning project using Python, supporting data processing pipelines, and experimental workflows.
- Built, refactored, and optimised automation workflows and Power Platform solutions (Power Automate, PowerShell, Power Apps) to streamline internal processes, improve performance and scalability, and reduce manual admin overhead.
- Supported organisational AI adoption by developing AI chatbots and automation workflows using Copilot Studio, integrating knowledge-grounded data (RAG) retrieval for agents, document processing, and custom AI model training.
- Assisted with internal AI enablement by helping staff adopt AI tools effectively and safely, promoting best practices around governance, security, and efficient usage.
- Provided helpdesk-level support, diagnosing, debugging and resolving production issues to improve system reliability and reduce downtime.

PROJECTS

Arduino Auth System (C++, Python, Javascript) | <https://github.com/joel06rob/Arduino-Auth-System>

- Developed an embedded authentication system integrating a web application, Python middleware, and Arduino microcontroller hardware
- Implemented serial communication protocols between host software and embedded hardware to transmit and display cryptographically generated authentication codes
- Designed and tested embedded state-management logic for code verification, user authentication, and hardware-driven feedback via an LED matrix display

STM32 and Arduino CAN Visualiser (C) | <https://github.com/joel06rob/STM32toArduino>

- Developed a bidirectional Classical CAN communication system between an STM32F446RE and Arduino Uno Q, with a browser-based interface for real-time CAN frame visualization.
- Implemented embedded CAN concepts including 500 kbit/s communication, CAN transceivers and bus termination, message filtering, interrupt/event-driven transmission and reception, UART debugging, and C++/Python/WebSocket integration.
- Built an end-to-end embedded-to-web debugging tool that decodes received CAN frames and reconstructs CANH/CANL dominant/recessive states, demonstrating practical understanding of CAN communication from the physical layer through to application-level visualization.

CEIR Terminal (C, Python) | <https://github.com/joel06rob/CEIR-Terminal>

- Developed a terminal-based vehicle diagnostics application in C, simulating multiple engine operating states and system behaviours
- Implemented sensor simulation modules for temperature, oil pressure, and diagnostic warnings, enabling real-time monitoring and fault detection scenarios
- Applied structured programming principles, modular code design, and debugging techniques to improve maintainability and reliability